Module Outline

Module Title	Information Risk Management		
Module Code	TBC	**New or revised**	New
NQF Level	7	**Credit Value**	10
JACS Code	TBC	**Duration (weeks)**	6
Date of Introduction	September 2020		
Module Supervisor	N/A		
Teaching Staff	Various		
Contact details for student enquiries	A dedicated student advisor will be appointed upon enrolment		
Campus(es) to be taught on	Online		
Terms in which module is taught	All		
Name of External Examiner:		**Existing external examiner**	**Y/N**
		If no, please state if nomination has been sent to QUAD	**Y/N**

Module Description

This module introduces students to the underpinning concepts and principles of Information Risk Management (IRM). This includes a review of traditional and contemporary Software Development Life Cycle (SDLC) models, focusing on the areas most affected by risk management considerations. The course will provide students with a combination of an understanding of IRM principles together with a review of assessment and mitigation techniques. Students will be introduced to the techniques in an engaging format, using a mixture of case studies, group work and individual activities.

Aims

The module aims to provide students with

1. An understanding of the basic principles of information risk management
2. An understanding of the relationship between IRM and the SDLC
3. An understanding of the role of risk management and residual risk in the SDLC
4. An appreciation for current and future challenges, limitations and opportunities.
5. The opportunity to reflect on and evaluate personal development

Learning Outcomes

1. Identify and analyse critically IT system risks and problems, and identify appropriate methodologies, tools and techniques to solve/mitigate them
2. Evaluate and adapt programs and systems to produce a solution that meets the design brief
3. Critically analyse and evaluate solutions produced
4. Systematically develop and implement the skills required to be effective member of a development team in a virtual professional environment, adopting real-life perspectives on roles and team organisation

Syllabus

- What is Information Risk Management (IRM)? IRM & the software development life cycle (SDLC) - sources of risk & standards
- Risk, business continuity and disaster recovery
- Systems evolution: the changing face of IT, risk and management.

Learning and Teaching Methods

The module will be delivered through the provision of specified reading materials on the virtual learning platform, which shall be supported by specified online discussion forums and lecturecasts. The lecturecasts and other learning materials will follow the best practices of 'diverse inclusion'. The tutor-led discussion forums will be available to the students in order to engage and enhance students' learning in a friendly and respectful environment. The flexible and participative approach of the module will develop a collaborative research inquiry in the advancement of computing, enabling them to accelerate in their chosen career.

Students will demonstrate their ability and strengths through evidence and reflections by maintaining an e-portfolio. The e-portfolio will also act as a means for assessment on evidence of personal growth and CPD. They will also identify and share appropriate web-based resources as learning support references with their fellow students and as indicators of their individual learning contexts with their tutors.

Synchronous sessions will last for one hour and allow students the opportunity for tutor contact and interaction with fellow students. The sessions will include live coding sessions, providing full demonstrations on the practical elements of the syllabus. They will also include referenced use of selected case studies which will be drawn from the reading materials/web-based module learning resources and the professional/educational experience of the Tutors. These synchronous sessions will be recorded in order to ensure that all students can access the material in their own time.

At pre-arranged days and agreed times during the module (usually weekly, prior to a synchronous session), the Module Tutor will be available for a 'drop in' telephone or online preparatory learning liaison session. This is to give students an opportunity to ask specific and general questions related to that week's learning opportunities and to enable them to contextualise their learning experience. Additionally, self-managed learning will supplement lectures and students are given weekly direction on required and indicative reading.

For team activities in this module, students will be grouped according to time zones to ensure team members can communicate easily with each other. They will then be encouraged to draw up a team contract, assigning roles and tasks to each team member to ensure they are able to achieve the given goal(s) for the project/assignment in the given timescale. There should be regular team meetings which will need to be recorded and logged. The final aspect of the group activity is the peer assessment. This will determine the final score of each team member for the project/assignment. Details on the process for peer assessment will be made available to students at the outset of the module.

<u>Key areas of Academic & Employability Skill Development in this module</u>

By the end of this module, students will have had the opportunity to develop skills in the following areas through summative and formative activities:

- Time management
- Commercial Awareness
- Critical thinking and analysis
- Communication and Literacy skills
- IT and Digital
- Numeracy
- Interpersonal
- Teamwork

<u>Assessment</u>

100% Coursework

Description of assessment for the website

<u>Formative Assessment</u>
To aid students' development of an in-depth understanding of the syllabus, regular formative assessment is provided via self-assessed multiple-choice questions, worksheets/exercises, a collaborative discussion forum and/or reflective commentary. Furthermore, the formative feedback received will enable students to develop their understanding of what is required for the summative assessments.

Formative feedback will also be given to several exercises the student will undertake as part of the risk assessment component of a project, working as a member of a project team. The output of these exercises will be open to peer and tutor review. The exercises will be based on techniques learned from recommended texts and lecturecasts and supported by discussions carried out via the forums and Synchronous sessions.

Formative feedback will also be provided on the module e-portfolio. This is a learning and development e-portfolio which should contain a summary of the module learning outcomes, relevant artefacts developed during the module along with a description for each one and their relation to the module learning outcomes, feedback from peers and tutors, team meeting notes and/or plans (where applicable), a reflective piece, professional skills matrix and action plan (PDP). Therefore, the e-portfolio will allow students to demonstrate their ability to reflect on their learning and personal development and create CPD action plans. The e-portfolio will need to be submitted at the end of the Unit 6 for formative assessment.

The tutor will also provide formative feedback on student participation in various activities in the module (such as the collaborative discussion forum), and on more skills-based elements of learning. This will be provided during the learning liaison sessions.

<u>Summative Assessment</u>
This will be based on the outcomes of the sessions above. The first assessment will be a project status report which is required half-way through the module. The team will then need to produce a final risk assessment report, which will include rationale and recommendations around the business continuity/ disaster recovery (BC/DR) strategy. Finally, students will need to produce an individual reflective piece on the skills and knowledge gained from the module and working as a member of a development team.

Assessment details

Description of unit of assessment (for exams, specify when the exam will be)	**Length / Duration**	**Submission Date**	**Weighting**	**Learning Outcome(s) met**
Development Team Project: Status Report	1 page (600 words equivalent)	Unit 3	30%	1, 2, 4
Development Team Project: Risk assessment report	1500 words equivalent	Unit 6	40%	1 – 4
Individual Reflective piece	500 words	Unit 6	30%	4

Re-assessment strategy (not for publication on the website)
Re-assessment will only be possible where permitted under the Rules of Assessment.
The standard KOL Variations to PG Rules of Assessment will apply. A maximum of 1 attempt at reassessment is allowed. Condonement may be granted subject to criteria specified in the PG Rules of Assessment.

Bibliography

Essential reading

- Sutton, D. (2015). *Information Risk Management*. 1st ed. Swindon, United Kingdom: BCS Learning & Development Limited.

Recommended Reading

- Stoneburner, G., Goguen, A. and Feringa, A. (2002). *SP 800-30. Risk Management Guide for Information Technology Systems*. [online] Dl.acm.org. Available at: https://dl.acm.org/citation.cfm?id=2206240 [Accessed 9 Dec. 2019].
- Spears, J. and Barki, H. (2010). User Participation in Information Systems Security Risk Management. *MIS Quarterly*, [online] 34(3), p.503. Available at: https://www.jstor.org/stable/25750689?seq=1 [Accessed 9 Dec. 2019].

List of courses in which the module is offered / to be offered, and module status - core, compulsory, optional)

MSc Cyber Security – Compulsory
PGDip Cyber Security – Compulsory

Other Information
Module assumes foundational knowledge of Software Development Life Cycle (SDLC) models.